

Unit 5: The Pythagorean Theorem

Practice Problems — All Four Sections

Name: _____ Date: _____ Period: _____

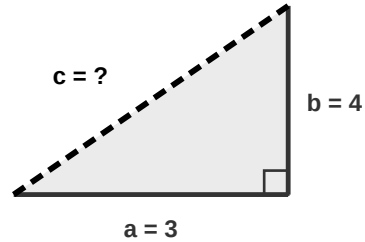
Key Formula: $a^2 + b^2 = c^2$ (a, b = legs; c = hypotenuse — always opposite the 90° angle)

5.1 — Introduction & How to Use

Determine if the missing side is a LEG or the HYPOTENUSE first, then solve.

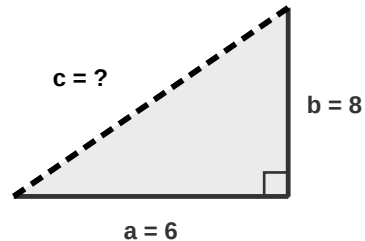
Part A — Solve for the Hypotenuse

Q1. A right triangle has legs 3 and 4. Find the hypotenuse.



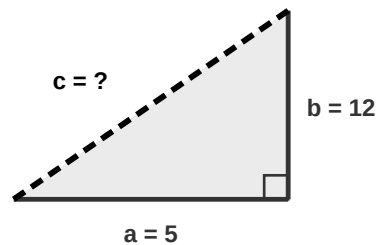
c =: _____

Q2. A right triangle has legs 6 and 8. Find the hypotenuse.



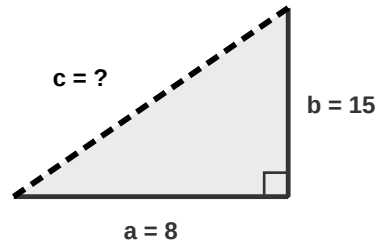
c =: _____

Q3. A right triangle has legs 5 and 12. Find the hypotenuse.



c =: _____

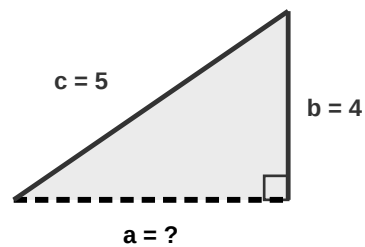
Q4. A right triangle has legs 8 and 15. Find the hypotenuse.



c =: _____

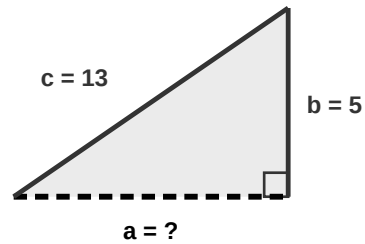
Part B — Solve for a Missing Leg

Q5. Hypotenuse = 5, leg = 4. Find the missing leg.



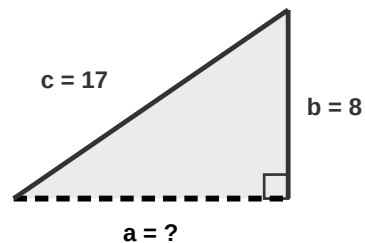
a =: _____

Q6. Hypotenuse = 13, leg = 5. Find the missing leg.



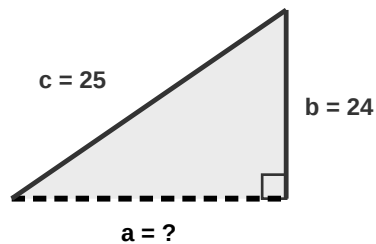
a =: _____

Q7. Hypotenuse = 17, leg = 8. Find the missing leg.



a =: _____

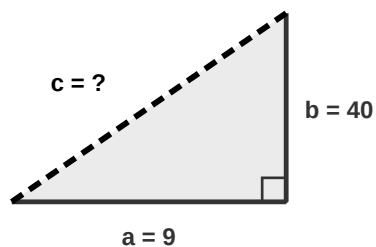
Q8. Hypotenuse = 25, leg = 24. Find the missing leg.



$a =$ _____

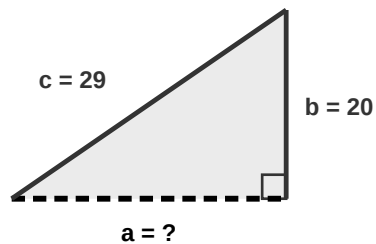
Part C — Identify the Missing Side, Then Solve

Q9. Legs are 9 and 40. Find the hypotenuse.



$c =$ _____

Q10. Hypotenuse = 29, leg = 20. Find the other leg.



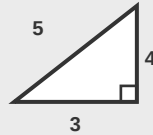
$a =$ _____

5.2 — Converse of the Pythagorean Theorem

Sort sides smallest to largest. Test: does $a^2 + b^2 = c^2$?

Part A — Classic Triples

Q1. Sides: 3, 4, 5. Is this a right triangle? Show your work.



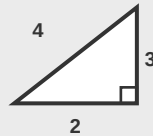
$$3^2 + 4^2 = \underline{\hspace{2cm}}$$

$$5^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Q2. Sides: 2, 3, 4. Is this a right triangle? Show your work.



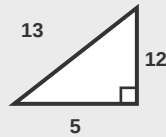
$$2^2 + 3^2 = \underline{\hspace{2cm}}$$

$$4^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Q3. Sides: 5, 12, 13. Is this a right triangle? Show your work.



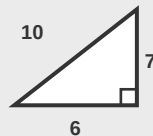
$$5^2 + 12^2 = \underline{\hspace{2cm}}$$

$$13^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Q4. Sides: 6, 7, 10. Is this a right triangle? Show your work.



$$6^2 + 7^2 = \underline{\hspace{2cm}}$$

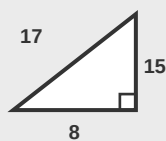
$$10^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Part B — Larger Numbers

Q5. Sides: 8, 15, 17. Is this a right triangle? Show your work.

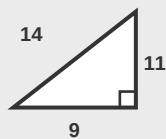


$$8^2 + 15^2 = \underline{\hspace{2cm}}$$
$$17^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Q6. Sides: 9, 11, 14. Is this a right triangle? Show your work.

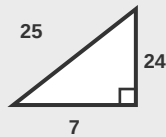


$$9^2 + 11^2 = \underline{\hspace{2cm}}$$
$$14^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Q7. Sides: 7, 24, 25. Is this a right triangle? Show your work.

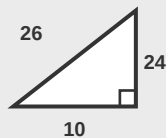


$$7^2 + 24^2 = \underline{\hspace{2cm}}$$
$$25^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Q8. Sides: 10, 24, 26. Is this a right triangle? Show your work.



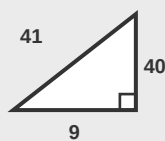
$$10^2 + 24^2 = \underline{\hspace{2cm}}$$
$$26^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Part C — Tricky Values

Q9. Sides: 9, 40, 41. Is this a right triangle? Show your work.



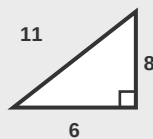
$$9^2 + 40^2 = \underline{\hspace{2cm}}$$

$$41^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

Yes / No: _____

Q10. Sides: 6, 8, 11. Is this a right triangle? Show your work. (Careful — is this the same as 6-8-10?)



$$6^2 + 8^2 = \underline{\hspace{2cm}}$$

$$11^2 = \underline{\hspace{2cm}}$$

Equal? Yes / No

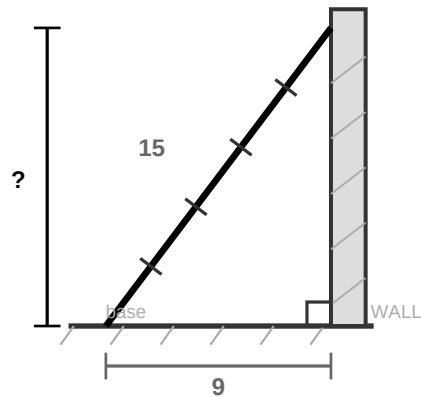
Yes / No: _____

5.3 — Word Problems

Label the diagram, identify the hypotenuse, then set up $a^2 + b^2 = c^2$ and solve.

Part A — Ladders, Ramps & Screens

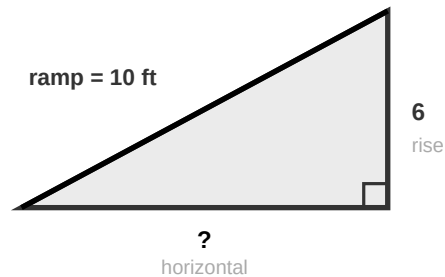
Q1. A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high up the wall does it reach?



Height on wall =: _____

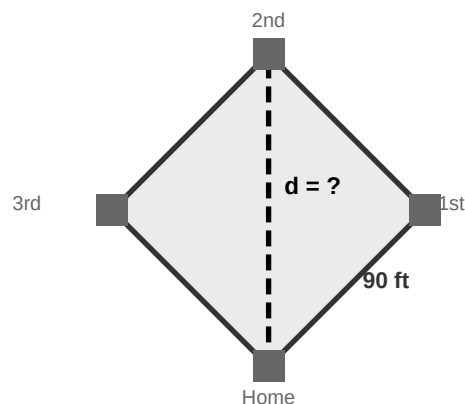
Wheelchair Ramp

Q2. A wheelchair ramp is 10 ft long and rises 6 ft vertically. What is the horizontal length?



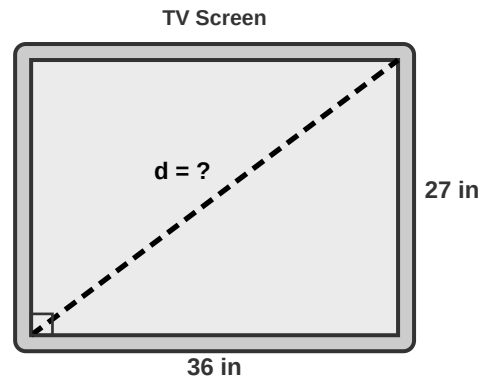
Horizontal length =: _____

Q3. A baseball diamond is a square with 90 ft sides. What is the distance from home plate to 2nd base? Round to the nearest foot.



Distance (nearest ft) =: _____

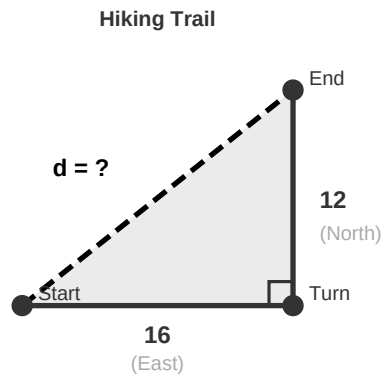
Q4. A TV screen is 36 in wide and 27 in tall. What is the diagonal measurement?



Diagonal =: _____

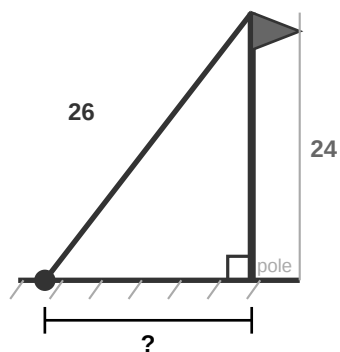
Part B — Navigation & Distance

Q5. A hiker walks 16 km East, then 12 km North. How far is she from the start in a straight line?



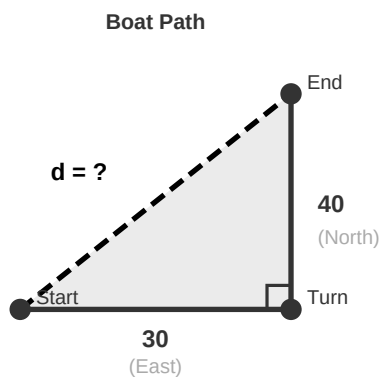
Straight-line distance =: _____

Q6. A flagpole is 24 ft tall. A wire runs from the top down to a stake in the ground. The wire is 26 ft long. How far is the stake from the base of the pole?



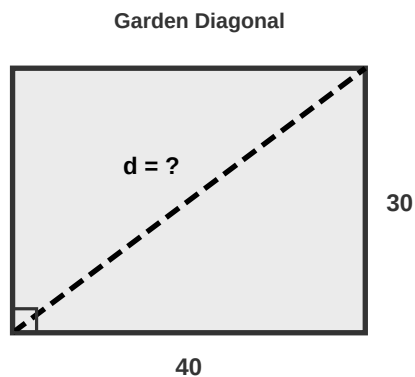
Distance from base =: _____

Q7. A boat travels 30 km East then 40 km North. How far is it from the dock in a straight line?



Straight-line distance =: _____

Q8. A rectangular garden is 40 m wide and 30 m long. How long is the diagonal path from one corner to the opposite corner?



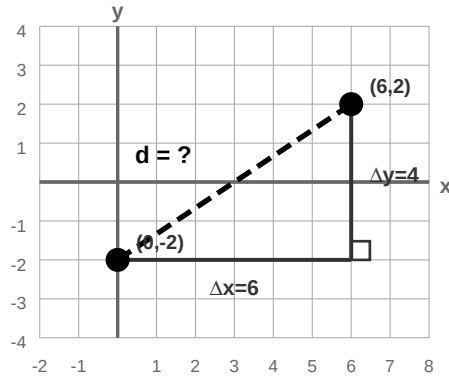
Diagonal =: _____

5.4 — Distance on the Coordinate Plane

Use the two points to form a right triangle, then apply the Pythagorean Theorem to find the distance.

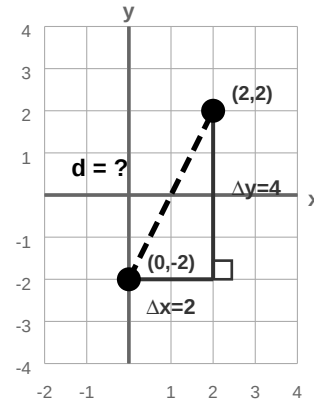
Part A — Integer Coordinates

Q1. Find the distance between (0, -2) and (6, 2). $\Delta x =$ _____ $\Delta y =$ _____



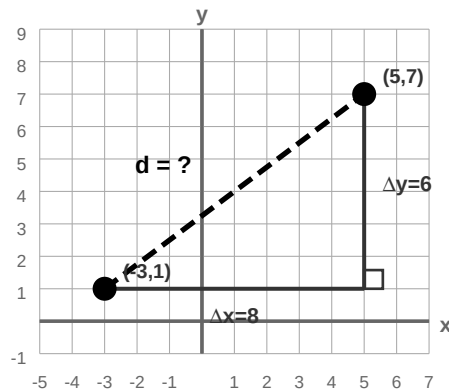
Distance: _____

Q2. Find the distance between (0, -2) and (2, 2). $\Delta x =$ _____ $\Delta y =$ _____



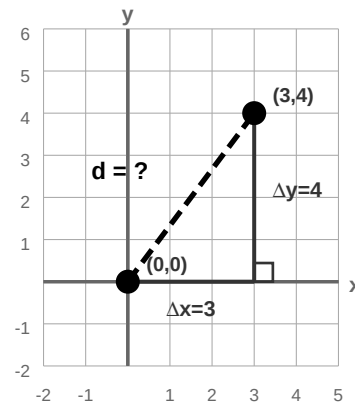
Distance: _____

Q3. Find the distance between (-3, 1) and (5, 7). $\Delta x =$ _____ $\Delta y =$ _____



Distance: _____

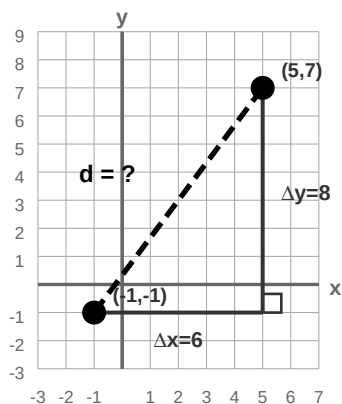
Q4. Find the distance between (0, 0) and (3, 4). $\Delta x =$ _____ $\Delta y =$ _____



Distance: _____

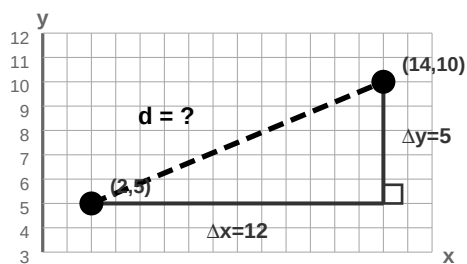
Part B — Negative Coordinates

Q5. Find the distance between $(-1, -1)$ and $(5, 7)$. $\Delta x =$ $\Delta y =$



Distance: _____

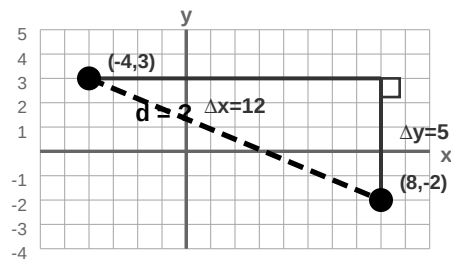
Q7. Find the distance between $(2, 5)$ and $(14, 10)$. Each unit = 1 mile. $\Delta x =$ $\Delta y =$



Distance: _____

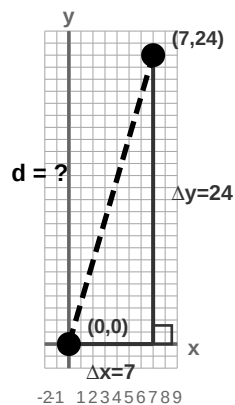
Part C — Applications

Q6. Find the distance between $(-4, 3)$ and $(8, -2)$. $\Delta x =$ $\Delta y =$



Distance: _____

Q8. Find the distance between $(0, 0)$ and $(7, 24)$. $\Delta x =$ $\Delta y =$



Distance: _____

Distance: _____

A coordinate plane with x and y axes ranging from -2 to 10. A line segment is drawn from the point $(0,0)$ to the point $(6,8)$. A right triangle is formed with the segment as the hypotenuse. The horizontal leg is labeled $\Delta x = 6$ and the vertical leg is labeled $\Delta y = 8$. A right angle symbol is shown at the vertex $(6,0)$. The distance d is marked with a question mark.

Distance: _____

ANSWER KEY

Unit 5: The Pythagorean Theorem — All Sections

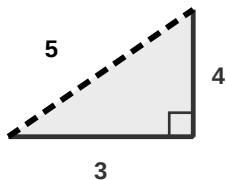
Section 5.1 — Introduction & How to Use

Q1. $c = 5$

$$3^2 + 4^2 = 9 + 16 = 25$$

$$c = \sqrt{25} = 5$$

→ 3-4-5 triple.

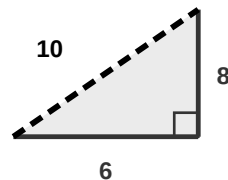


Q2. $c = 10$

$$6^2 + 8^2 = 36 + 64 = 100$$

$$c = \sqrt{100} = 10$$

→ 3-4-5 $\times 2$.

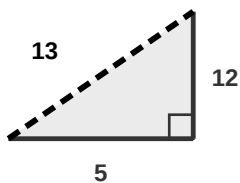


Q3. $c = 13$

$$5^2 + 12^2 = 25 + 144 = 169$$

$$c = \sqrt{169} = 13$$

→ 5-12-13 triple.

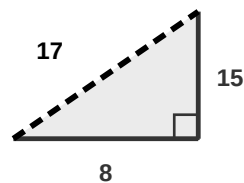


Q4. $c = 17$

$$8^2 + 15^2 = 64 + 225 = 289$$

$$c = \sqrt{289} = 17$$

→ 8-15-17 triple.



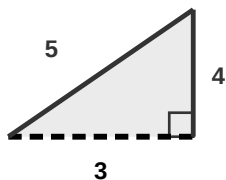
Q5. $a = 3$

$$a^2 + 4^2 = 5^2$$

$$a^2 = 25 - 16 = 9$$

$$a = \sqrt{9} = 3$$

→ Subtract leg^2 from hyp^2 .



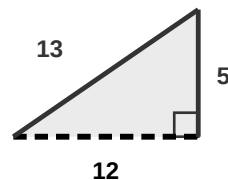
Q6. $a = 12$

$$a^2 + 5^2 = 13^2$$

$$a^2 = 169 - 25 = 144$$

$$a = \sqrt{144} = 12$$

→ 5-12-13 triple.



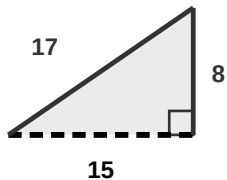
Q7. a = 15

$$a^2 + 8^2 = 17^2$$

$$a^2 = 289 - 64 = 225$$

$$a = \sqrt{225} = 15$$

→ 8-15-17 triple.



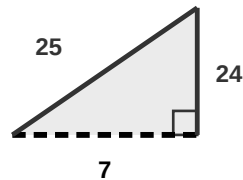
Q8. a = 7

$$a^2 + 24^2 = 25^2$$

$$a^2 = 625 - 576 = 49$$

$$a = \sqrt{49} = 7$$

→ 7-24-25 triple.

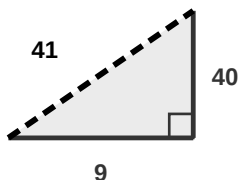


Q9. c = 41

$$9^2 + 40^2 = 81 + 1600 = 1681$$

$$c = \sqrt{1681} = 41$$

→ 9-40-41 triple.



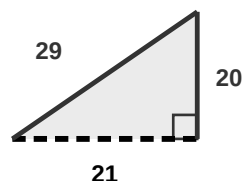
Q10. a = 21

$$a^2 + 20^2 = 29^2$$

$$a^2 = 841 - 400 = 441$$

$$a = \sqrt{441} = 21$$

→ 20-21-29 triple.

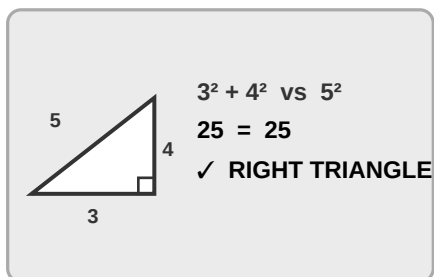


Section 5.2 — Converse

Q1. YES

$$3^2 + 4^2 = 25, \quad 5^2 = 25 \quad \checkmark$$

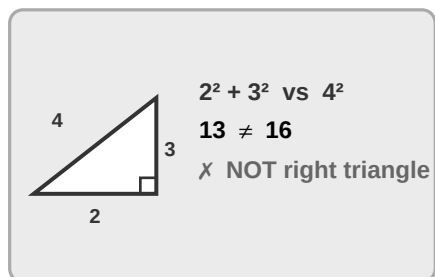
→ 3-4-5 triple.



Q2. NO

$$2^2 + 3^2 = 13, \quad 4^2 = 16 \quad \times \quad (13 \neq 16)$$

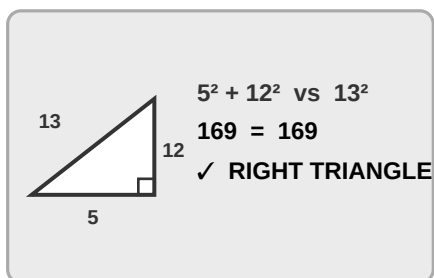
→ Not a triple.



Q3. YES

$$5^2 + 12^2 = 169, \quad 13^2 = 169 \quad \checkmark$$

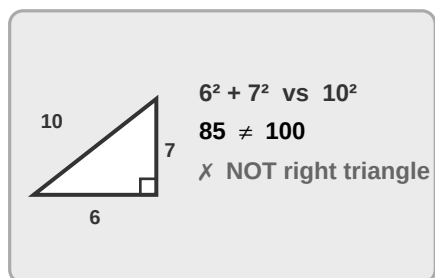
→ 5-12-13 triple.



Q4. NO

$$6^2 + 7^2 = 85, \quad 10^2 = 100 \quad \times$$

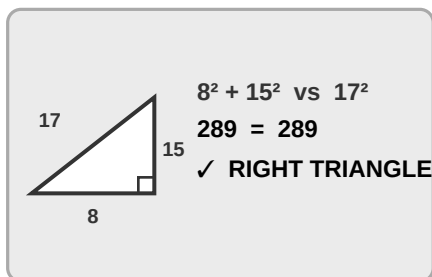
→ $85 \neq 100$.



Q5. YES

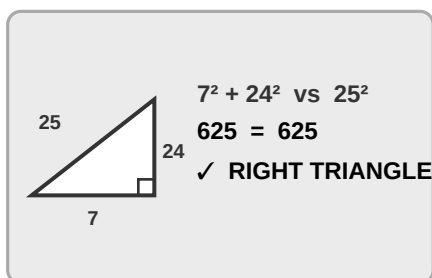
$$8^2 + 15^2 = 289, \quad 17^2 = 289 \quad \checkmark$$

→ 8-15-17 triple.

**Q7. YES**

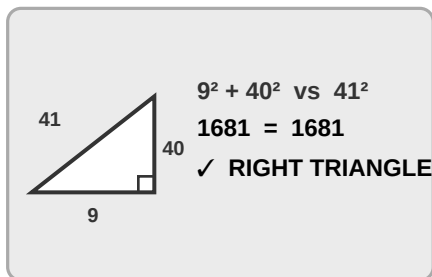
$$7^2 + 24^2 = 625, \quad 25^2 = 625 \quad \checkmark$$

→ 7-24-25 triple.

**Q9. YES**

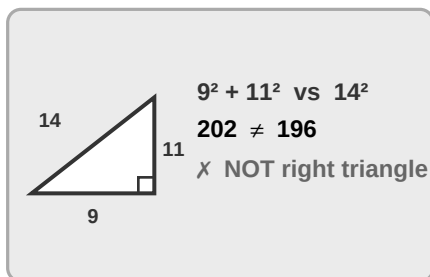
$$9^2 + 40^2 = 1681, \quad 41^2 = 1681 \quad \checkmark$$

→ 9-40-41 triple.

**Q6. NO**

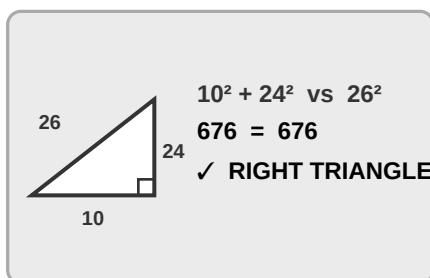
$$9^2 + 11^2 = 202, \quad 14^2 = 196 \quad \times$$

→ 202 ≠ 196.

**Q8. YES**

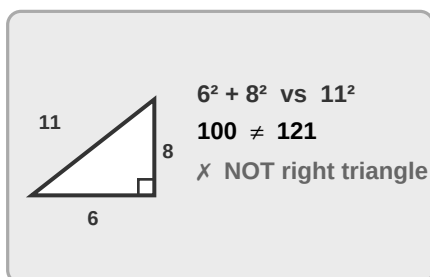
$$10^2 + 24^2 = 676, \quad 26^2 = 676 \quad \checkmark$$

→ 5-12-13 ×2.

**Q10. NO**

$$6^2 + 8^2 = 100, \quad 11^2 = 121 \quad \times$$

(6-8-10 works, not 6-8-11!)
→ Trick!

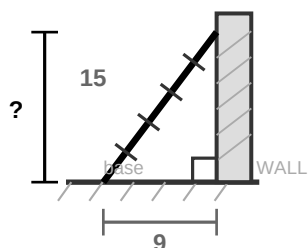
**Section 5.3 — Word Problems****Q1. h = 12 ft**

$$9^2 + h^2 = 15^2$$

$$81 + h^2 = 225$$

$$h^2 = 144 \rightarrow h = 12$$

→ 3-4-5×3. Ladder=hyp.

**Q2. 8 ft**

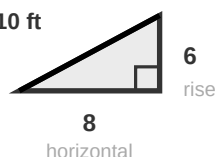
$$a^2 + 6^2 = 10^2$$

$$a^2 = 64 \rightarrow a = 8$$

→ 6-8-10 triple.

Wheelchair Ramp

ramp = 10 ft

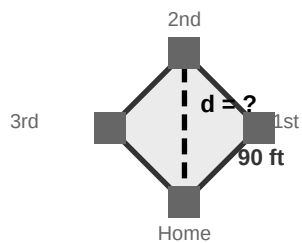


Q3. ≈ 127 ft

$$d^2 = 90^2 + 90^2 = 16200$$

$$d = 90\sqrt{2} \approx 90 \times 1.414 \approx 127.3$$

→ Square diagonal = $s\sqrt{2}$.

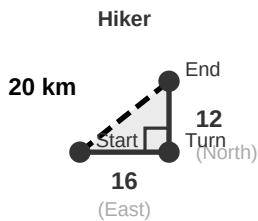


Q5. 20 km

$$16^2 + 12^2 = 256 + 144 = 400$$

$$d = \sqrt{400} = 20$$

→ 3-4-5 $\times 4$.

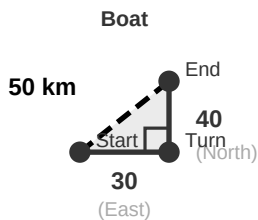


Q7. 50 km

$$30^2 + 40^2 = 900 + 1600 = 2500$$

$$d = \sqrt{2500} = 50$$

→ 3-4-5 $\times 10$.

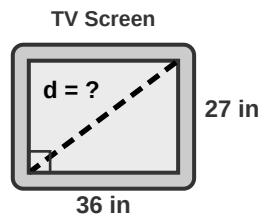


Q4. 45 in

$$36^2 + 27^2 = 1296 + 729 = 2025$$

$$d = \sqrt{2025} = 45$$

→ 3-4-5 $\times 9$.

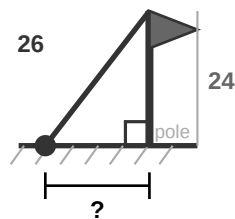


Q6. 10 ft

$$a^2 + 24^2 = 26^2$$

$$a^2 = 676 - 576 = 100 \rightarrow a = 10$$

→ 5-12-13 $\times 2$.

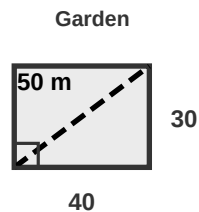


Q8. 50 m

$$40^2 + 30^2 = 1600 + 900 = 2500$$

$$d = \sqrt{2500} = 50$$

→ 3-4-5 $\times 10$.



Section 5.4 — Coordinate Plane Distance

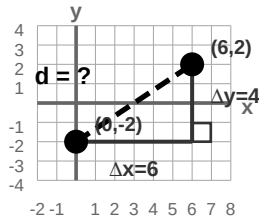
Q1. $d = \sqrt{52} \approx 7.21$

$$\Delta x = 6, \Delta y = 4$$

$$d^2 = 36 + 16 = 52$$

$$d = \sqrt{52} = 2\sqrt{13} \approx 7.21$$

→ Irrational — no whole number answer.



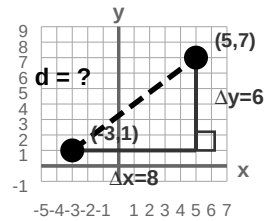
Q3. $d = 10$

$$\Delta x = 8, \Delta y = 6$$

$$d^2 = 64 + 36 = 100$$

$$d = 10$$

→ 6-8-10 triple!



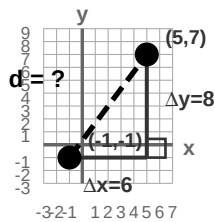
Q5. $d = 10$

$$\Delta x = 5 - (-1) = 6, \Delta y = 7 - (-1) = 8$$

$$d^2 = 36 + 64 = 100$$

$$d = 10$$

→ Subtract negatives carefully. 6-8-10 triple.



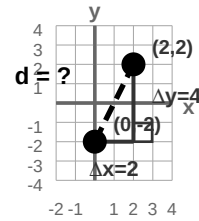
Q2. $d = \sqrt{20} \approx 4.47$

$$\Delta x = 2, \Delta y = 4$$

$$d^2 = 4 + 16 = 20$$

$$d = \sqrt{20} = 2\sqrt{5} \approx 4.47$$

→ Irrational.



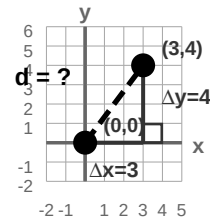
Q4. $d = 5$

$$\Delta x = 3, \Delta y = 4$$

$$d^2 = 9 + 16 = 25$$

$$d = 5$$

→ 3-4-5 triple.



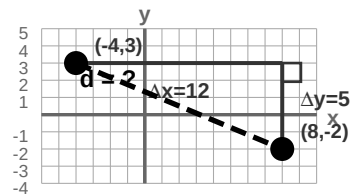
Q6. $d = 13$

$$\Delta x = 8 - (-4) = 12, \Delta y = |-2 - 3| = 5$$

$$d^2 = 144 + 25 = 169$$

$$d = 13$$

→ 5-12-13 triple.



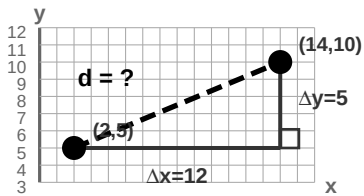
Q7. 13 miles

$$\Delta x=12, \Delta y=5$$

$$d^2=144+25=169$$

$$d=13 \text{ miles}$$

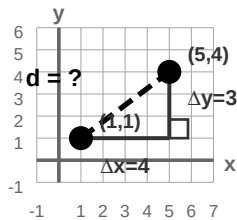
→ 5-12-13 triple.

**Q9. AC = 5**

$$AB=4 \text{ (horiz)}, BC=3 \text{ (vert)}$$

$$AC^2=16+9=25 \rightarrow AC=5$$

→ 3-4-5 triple.

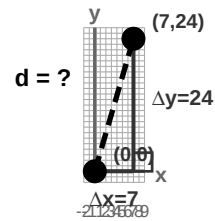
**Q8. d = 25**

$$\Delta x=7, \Delta y=24$$

$$d^2=49+576=625$$

$$d=25$$

→ 7-24-25 triple.

**Q10. QR = 10**

$$PQ=6, PR=8$$

$$QR^2=36+64=100 \rightarrow QR=10$$

→ Right angle at P(0,0).

